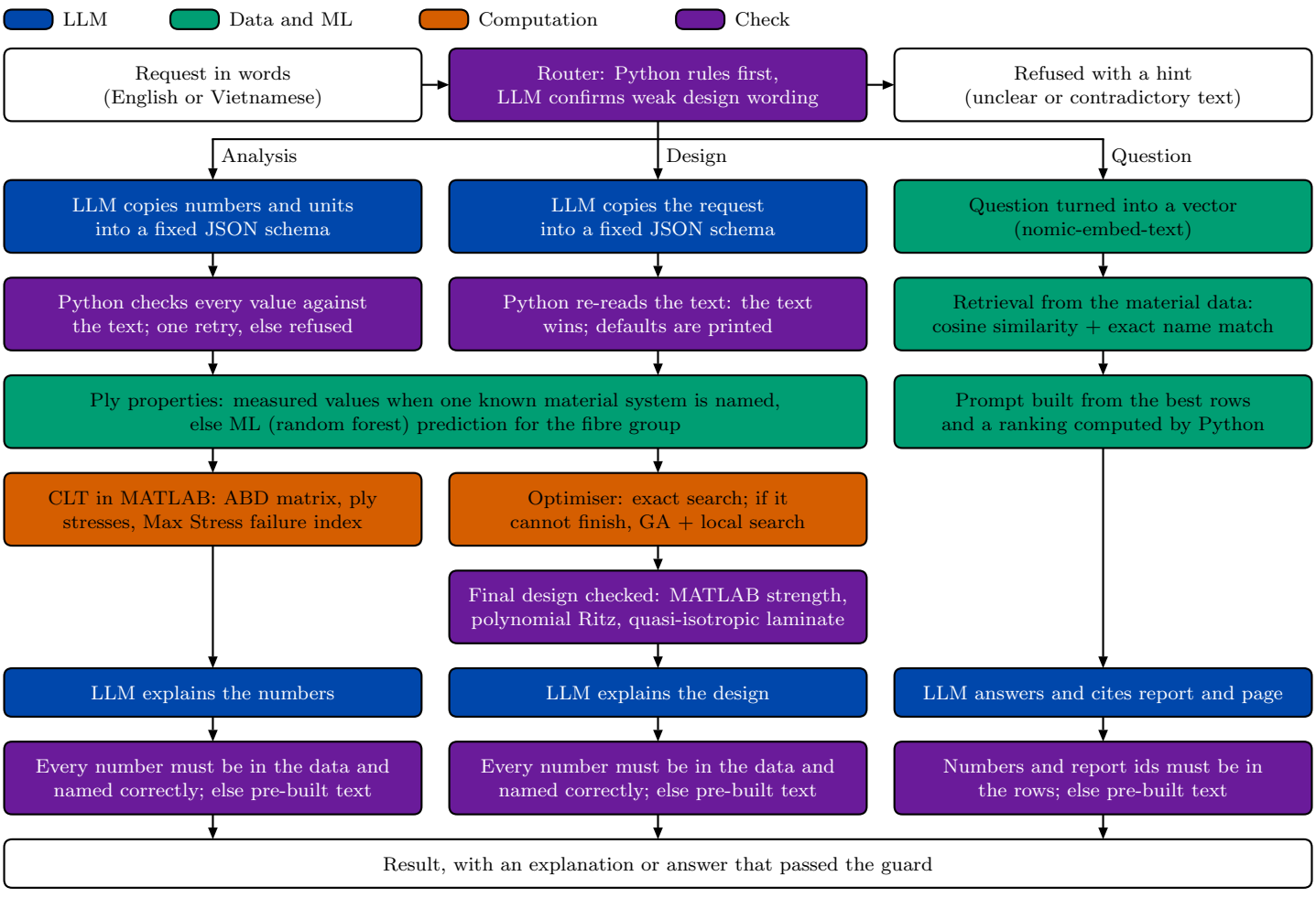
LLM
Data and ML
Computation
Check
Request in words
(English or Vietnamese)
Router: Python rules first,
LLM confirms weak design wording
Refused with a hint
(unclear or contradictory text)
Analysis
Design
Question
LLM copies numbers and units
into a fixed JSON schema
LLM copies the request
into a fixed JSON schema
Question turned into a vector
(nomic-embed-text)
Python checks every value against
the text; one retry, else refused
Python re-reads the text: the text
wins; defaults are printed
Retrieval from the material data:
cosine similarity + exact name match
Ply properties: measured values when one known material system is named,
else ML (random forest) prediction for the fibre group
Prompt built from the best rows
and a ranking computed by Python
CLT in MATLAB: ABD matrix, ply
stresses, Max Stress failure index
Optimiser: exact search; if it
cannot finish, GA + local search
Final design checked: MATLAB strength,
polynomial Ritz, quasi-isotropic laminate
LLM explains the numbers
LLM explains the design
LLM answers and cites report and page
Every number must be in the data and
named correctly; else pre-built text
Every number must be in the data and
named correctly; else pre-built text
Numbers and report ids must be in
the rows; else pre-built text
Result, with an explanation or answer that passed the guard